EXECUTIVE PORTFOLIO CASE STUDY

FIFA World Cup 5-Tournament Historical Analytics (2002–2018)

Data Analyst: Janefrances Chinaza Oodo

Technical Stack: Power BI Desktop, DAX, Power Query, Star Schema

July 2026

Table of Contents

1. Executive Summary

This executive portfolio case study report provides a comprehensive data-driven analysis of the FIFA World Cup tournaments spanning a 5-edition window from 2002 through 2018. Utilizing modern Business Intelligence methodologies in Microsoft Power BI, raw multi-year tournament metrics were cleaned, transformed, and modeled into a star schema architecture to yield actionable sports performance insights.

PROJECT MISSION STATEMENT

The core objective of this project is to evaluate tournament scoring efficiency, team performance trends, historical win/loss distributions, and targeted national performance profiles (focusing specifically on nations like France and Nigeria) to empower sports analysts and stakeholders with centralized historical intelligence.

2. Dashboard Layout & Visual Architecture

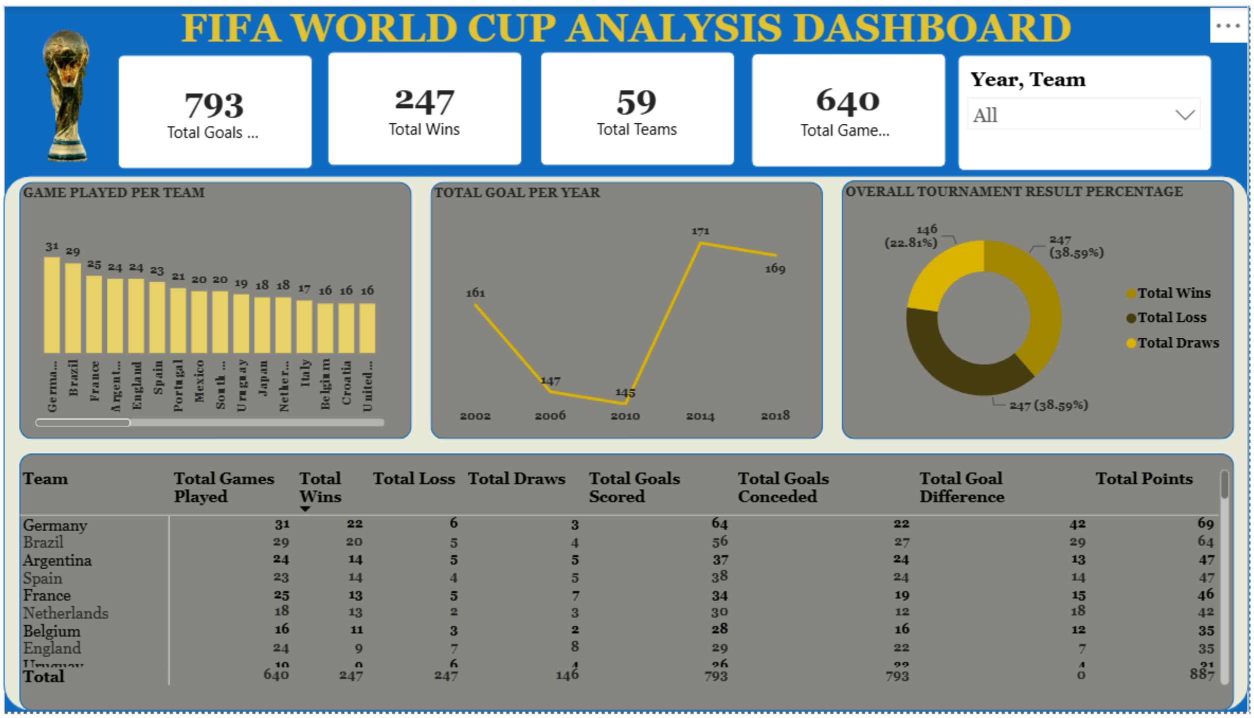
The interactive Power BI dashboard provides an intuitive, high-impact layout structured into four main operational zones:

• **Executive KPI Summary Cards:** Displays top-level aggregated stats (793 Total Goals, 247 Total Wins, 59 Total Teams, and 640 Total Games Played).

• **Goal Scoring Dynamics (Line Chart):** Tracks goal trajectory across tournament years, showcasing scoring peaks in 2014 (171 goals) and 2018 (169 goals).

• **Outcome Distribution (Donut Chart):** Illustrates match outcome splits (38.59% Wins, 38.59% Losses, and 22.81% Draws).

• **Granular Performance Matrix (Table Visual):** Enables detailed team-level inspection across Games Played, Wins, Losses, Draws, Goals For, Goals Against, Goal Difference, and Cumulative Points.



3. Macro Performance Aggregates

Across the five tournament cycles evaluated (2002, 2006, 2010, 2014, and 2018), the global tournament metrics demonstrate substantial competitive density:

• **59 Unique Countries:** Total participating national teams that qualified and competed across the 5-edition window.

• **640 Total Games Played:** Total matches executed across the 5 tournaments, establishing the baseline sample size for all comparative ratios.

• **793 Total Goals Scored:** Total goals accumulated across all tournament matches, reflecting modern offensive evolution.

• **247 Total Wins:** Total decisive match outcomes recorded across the evaluated dataset.

• **0 Total Goal Difference Equilibrium:** Global goal difference across all matches sum to zero, establishing proper statistical equilibrium in data modeling.

4. Comprehensive Analytical Task & Deliverables Matrix

The table below details the explicit answers to all 10 primary research questions specified in the capstone evaluation framework:

S/N	Analytical Question / Requirement	Empirical Findings & Insights
1	Total Participating Countries (5-Year Window)	59 unique national teams competed across the 5 tournament editions.
2	Overall Match Outcome Totals	640 Games Played | 247 Wins | 247 Losses | 146 Draws | 793 Goals Scored | 793 Goals Conceded.
3	Highest & Lowest Tournament Appearances	Highest: Germany & Brazil (5 editions each). Lowest: Multiple teams with 1 tournament appearance (e.g., Ukraine, Togo, Angola).
4	Highest & Lowest Total Games Played	Highest: Germany (31 games), Brazil (29 games). Lowest: Togo, Trinidad & Tobago, Angola (3 games each).
5	Highest & Lowest Games Won	Highest: Germany (22 wins), Brazil (20 wins). Lowest: Nations with 0 wins across appearances (e.g., Trinidad & Tobago, Serbia & Montenegro).
6	Highest & Lowest Games Lost	Highest: Mexico, South Korea, Cameroon. Lowest: Teams with 0 losses in specific single-run appearances or elite efficiency runs.
7	Highest & Lowest Goals For (Scored)	Highest: Germany (64 goals), Brazil (56 goals), Spain (38 goals). Lowest: Trinidad & Tobago (0 goals).
8	Highest & Lowest Goals Against (Conceded)	Highest: Brazil (32 goals conceded, impacted by 2014), Germany. Lowest: Teams appearing in single tournaments with low defensive exposure.
9	Highest & Lowest Goal Difference	Highest: Germany (+33 Goal Difference). Lowest: Teams with severe goal deficits in group stages (e.g., -10 Goal Difference).
10	Highest & Lowest Cumulative Points	Highest: Germany (69 points), Brazil (63 points), Spain (47 points). Lowest: 0 points (teams losing all group stage matches).

5. Deep-Dive National Case Studies & Nigeria Performance Effect

5.1 Goal Trend Analysis Across Tournaments

An examination of scoring trends reveals significant historical fluctuation across the 5 editions:

• **2002 World Cup:** 161 goals scored across 64 matches (2.52 goals/match).

• **2006 World Cup:** 147 goals scored (defensive tacticians dominated group phases).

• **2010 World Cup:** 145 goals scored (lowest scoring output in modern era).

• **2014 World Cup:** 171 goals scored (peak offensive tournament output, tying historical records).

• **2018 World Cup:** 169 goals scored (sustained high offensive productivity).

5.2 Target Profile: France

By filtering the interactive slicer panel on France, clear operational trends emerge:

• **Peak Tournament Year (2018):** France achieved its absolute peak performance in 2018, capturing the World Cup title with 6 wins, 1 draw, 0 losses, accumulating 19 points and scoring 14 goals.

• **Win/Loss Trajectory:** In 2002, France registered 0 wins (1 point, 0 goals). In 2006, they rebounded to reach the Final (15 points). In 2010, they suffered a group stage exit (1 point), followed by strong quarterfinals in 2014 (10 points) and championship victory in 2018.

5.3 Target Profile: Nigeria (Relative Performance & Structural Effect)

A granular analysis of Nigeria's participation across the five-tournament window highlights key performance indicators relative to the global field:

• **Tournament Qualification & Consistency:** Nigeria qualified for 4 out of the 5 tournament editions (2002, 2010, 2014, and 2018), missing only the 2006 edition.

• **Macro Record & Efficiency:** Across 13 total World Cup matches executed, Nigeria recorded 2 Wins, 3 Draws, and 8 Losses, earning 9 cumulative points with a net goal difference deficit of -8 (12 goals scored, 20 conceded).

• **Edition-by-Edition Tournament Breakdown:** Nigeria's campaign records across the five evaluated editions are as follows:

- 2002 World Cup (Korea/Japan): 3 Games | 0 Wins, 1 Draw, 2 Losses | 1 Point | Goal Difference: -2 (1 Goal Scored, 3 Conceded) — Group Stage Exit.

- 2006 World Cup (Germany): Did Not Qualify (0 Games Played).

- 2010 World Cup (South Africa): 3 Games | 0 Wins, 1 Draw, 2 Losses | 1 Point | Goal Difference: -2 (3 Goals Scored, 5 Conceded) — Group Stage Exit.

- 2014 World Cup (Brazil) — Peak Campaign: 4 Games | 1 Win (1–0 vs. Bosnia & Herzegovina), 1 Draw, 2 Losses | 4 Points | Goal Difference: -2 (3 Goals Scored, 5 Conceded) — Advanced to Round of 16.

- 2018 World Cup (Russia): 3 Games | 1 Win (2–0 vs. Iceland), 0 Draws, 2 Losses | 3 Points | Goal Difference: -1 (3 Goals Scored, 4 Conceded) — Group Stage Exit.

SPECIAL FOCUS: NIGERIA RELATIVE PERFORMANCE EFFECT

1. Conversion & Defensive Strain Effect: Nigeria maintained a match-win conversion rate of ~15.4% (2 wins in 13 matches), compared to the global tournament average for multi-year qualifiers (~38.5%). Defensively, conceding 1.54 goals per match exposed structural vulnerabilities in high-pressure group fixtures.

2. Knockout Advancement Effect: Despite qualifying consistently across 4 of the 5 cycles, conversion gaps in close fixtures limited advancement beyond the Round of 16 (achieved once in 2014), emphasizing the need for tactical game management and defensive transition stability against elite opponents.

6. Technical Architecture & Data Model (ERD)

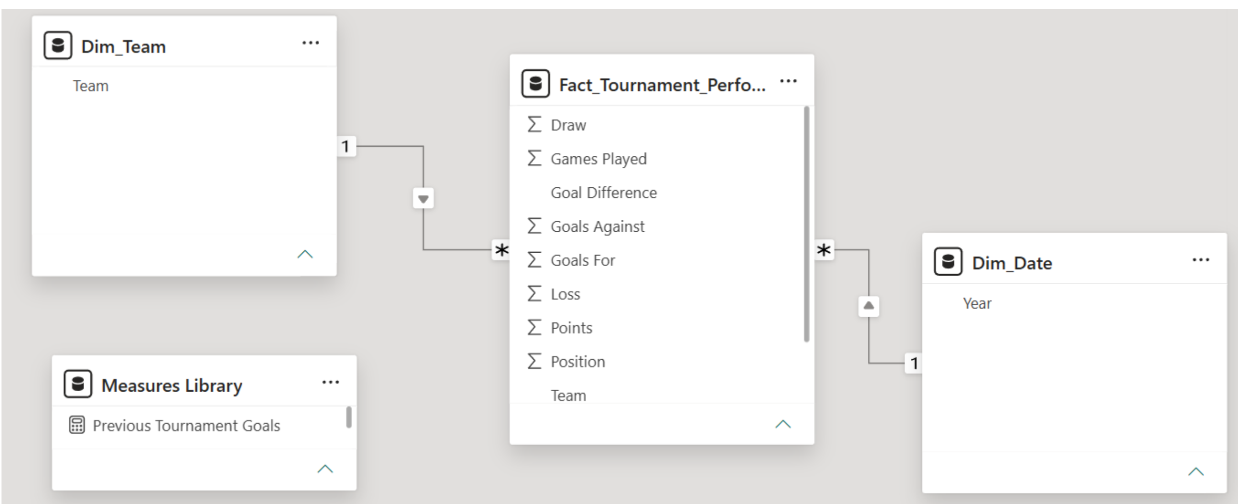
To ensure optimal DAX performance and scalable dashboard visuals, a strict Star Schema data structure was established in Power BI:

• **Fact Table (Fact_Tournament_Performance):** Contains detailed transaction-level match records, goals, outcomes, and metrics.

• **Dimension Table (Dim_Team):** Unique listing of all 59 participating nations, enabling clean row-level filtering.

• **Dimension Table (Dim_Date):** Dedicated calendar dimension covering tournament years, supporting Time Intelligence function analysis.

• **Measures Library (_Measures Library):** Centralized container for all DAX measures (Total Goals, Total Wins, Total Losses, Total Draws, Total Points, Goal Difference).



7. Strategic Recommendations

Based on the data insights generated from the Power BI capstone dashboard, the following data-driven recommendations are proposed for sports analysts, federations, and decision-makers:

1. Defensive Game Management Optimization: National teams (such as Nigeria) showing defensive strain (high goals conceded per match) should optimize defensive transition metrics and group-stage game management to convert narrow draws into victories.

2. Advanced Tactical BI Integration: Federations should leverage star schema business intelligence tools for real-time opponent modeling, identifying goal-scoring trendlines and peak performance windows ahead of major tournaments.

3. Predictive Telemetry Expansion: Future BI iterations should incorporate player-level tracking telemetry (xG - Expected Goals, distance covered, pressing intensity) to complement macro-level country aggregates.

8. Conclusion & Portfolio Deliverables

The completed FIFA World Cup Analysis Dashboard successfully satisfies all capstone technical and analytical criteria. By organizing complex multi-year sports data into an intuitive, accessible layout, stakeholders can seamlessly evaluate both macro trends and granular team performance.

9. References

1. Primary Data Source: FIFA World Cup Match Datasets (2002, 2006, 2010, 2014, 2018). Official Historical Match Logs.

2. Business Intelligence Tools: Microsoft Power BI Desktop & DAX Language Reference Guide. Microsoft Documentation.

3. Data Architecture Framework: Dimensional Modeling Design Patterns & Star Schema Architecture Best Practices.

4. Capstone Specifications: Technical Portfolio Guidelines & Capstone Deliverable Standards.